Proposal: Task Scheduling Application

1. Introduction

This proposal outlines the plan for the development of a comprehensive task scheduling application using the integrated AI capabilities of Replit. This approach leverages Replit's full-stack development environment and AI-powered coding assistance to build a robust and efficient application. The app will provide a centralized platform for users to manage and track tasks, with distinct functionalities for both administrators and regular users.

2. Project Goals & Objectives

The primary objective of this project is to create a user-friendly and reliable task scheduling app with the following key features, built efficiently with Replit AI:

- **Intuitive Interface:** A clean and easy-to-navigate user interface for seamless task management.
- **Dual User Roles:** The application will support two distinct user types: **Admin** and **User**, each with unique permissions and capabilities.
- **Secure Access:** A robust login and registration system to ensure data security.
- **Task Management:** The ability for users to manage tasks assigned to them, and for admins to oversee and assign tasks to their team.
- **Cross-Platform Accessibility:** A responsive design that works flawlessly on desktop and mobile browsers.

3. Core Features & Functionality

The application will be built around two primary user roles, each with a specific set of functionalities.

User-Facing Features

- **Registration:** Users can register by creating a username, password, and entering a unique Admin ID to link their account.
- **Login:** Users can log in using their username and password.
- **Task View:** Users can see a calendar view of all tasks assigned to them by their admin, including the task name and deadline.
- **Task Completion:**
  - Users can mark a task as finished using a "Finish" checkbox.
  - Users can upload evidence of completion, which includes a brief text description and the ability to upload documents, images, or videos.
- **Status Tracking:** The status of a task will change after the Admin verifies the submitted evidence and marks it as complete, which will be indicated by an "Admin Seen" checkbox.

Admin-Facing Features

- **Registration:** Admins can register by creating a username, password, and a unique Admin ID for their organization.
- **Login:** Admins can log in using their username and password.
- **User Management:** Admins can see a list of all users who have registered with their unique Admin ID.
- **Task Assignment:** Admins can schedule work for their team members by providing the following inputs:
  - Task Name
  - User Name (to assign the task to a specific user)
  - Deadline
- **Evidence Review:** Admins can review the evidence submitted by a user for a task.
- **Task Completion & Remarks:**
  - Admins can provide remarks on the evidence.
  - Admins can mark a task as completed using a "Completed" checkbox after verifying the evidence. This action will be reflected on the user's task view.

4. Scope of Work

The project will be completed in two main phases, leveraging the AI-powered development environment for rapid creation:

Phase 1: Planning & AI-Guided Development

- **Requirement Gathering:** Detailed discussions to finalize all features and functionality.
- **Prompt-Based Generation:** Using detailed prompts to guide Replit AI in generating the initial codebase, including the UI, backend logic, and database setup.
- **Collaborative Coding:** Working with the AI to iteratively refine and build out the application's features.

Phase 2: Refinement & Deployment

- **Quality Assurance (QA) Testing:** Rigorous testing to identify and fix any issues, ensuring a bug-free application.
- **Deployment:** Launching the application directly from the Replit environment.
- **Post-Launch Support:** Providing ongoing maintenance and support.

5. Technology Stack

The entire application will be built and hosted within the **Replit** platform, utilizing its integrated tools for full-stack development, including:

- **Replit AI:** For generating and assisting with code.
- **Frontend:** HTML, CSS, and JavaScript (or a framework like React).
- **Backend:** Node.js or a similar server-side environment.